

and Delta. Then, create several if conditions for selecting the appropriate servo motor that is attached to various joints on the prosthetic/robotic arm.

For example, to move the little finger of the prosthetic/robotic hand with brain signals, concentrate on the servo motor attached to it. When the preset threshold value

that is present in the code gets met, the servo motor will move and consequently move the little finger. Likewise, the same can be tried with rest of the fingers and the wrist.

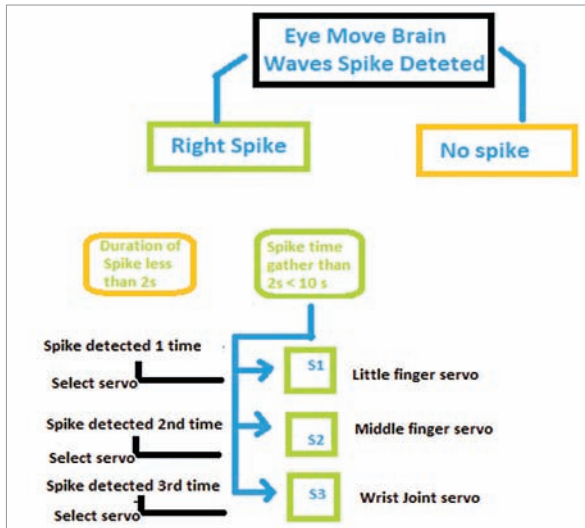


Fig. 8: Algorithm

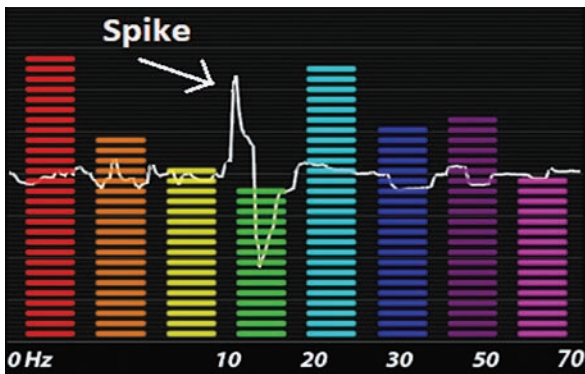


Fig. 9: Spike in graph observed during eye blink in EEC sensor reading

```

pi4 (raspberrypi) - VNC Viewer
*Python 3.7.3 Shell*
File Edit Shell Debug Options Window Help
atdeb=54
Moving robot wrist position
atdeb=54
Moving robot wrist position
atdeb=67
Moving robot wrist
atdeb=67
Moving robot wrist
atdeb=67
Moving robot wrist
atdeb=67
Moving robot wrist
atdeb=78
Moving robot wrist
atdeb=78
Moving robot wrist
atdeb=78
Moving robot wrist
atdeb=78
Moving robot wrist
atdeb=94
Moving robot wrist
atdeb=94
Moving robot wrist
atdeb=94
Moving robot wrist
atdeb=94
Moving robot wrist
atdeb=94
Moving robot wrist
blink
Servo select signal.....
4
atdeb=87
Moving robot thumb
  
```

Fig. 11: Code testing



Fig. 10: Another view of the spike in graph observed during eye blink in EEC sensor reading

Algorithm for selecting finger movement using brain waves

The selection of correct joint movement of a wrist or fingers in a prosthetic/robotic hand using the various brain waves is a bit complicated to understand. So, I am briefly explaining below how it works.

The brain signals generally fluctuate whenever we try to close and open our eyes. These fluctuations result in a spike in the brain signals that fully and properly control the blinking of the eyes. By leveraging this spike, you can select and accordingly move